

Characterizing Electricity Demand and Marginal CO₂ Emissions During Hazardous Grid Events

A Data-Mining Study of Grid Resilience and Carbon Intensity in the California ISO, 2019–2026

Pratik Patil
University of Colorado Boulder
Data Science
Boulder, CO, USA
pratik.patil@colorado.edu

Atharva Zodpe
University of Colorado Boulder
Data Science
Boulder, CO, USA
atharva.zodpe@colorado.edu

Moulik Kumar
University of Colorado Boulder
Data Science
Boulder, CO, USA
moulik.kumar@colorado.edu

Abstract

Climate-driven hazardous grid events — Public Safety Power Shut-offs (PSPS), extreme heat emergencies, and equipment failures — are reshaping electricity demand and marginal carbon emissions in ways that standard annual reporting cannot capture. This report presents a complete data-mining study of how such events alter the coupled dynamics of electricity demand and marginal CO₂ emission rates (MOER) across the California Independent System Operator (CAISO).

We integrate 5–15-minute MOER readings from the WattTime API with hourly demand from the U.S. Energy Information Administration’s EIA-930 platform, covering a curated catalog of 52 documented events between 2019 and 2026. Four complementary models are implemented: (1) **K-Means Clustering** to discover operational grid-state archetypes from unlabeled time-series data; (2) **Logistic Regression** to predict whether an automatically detected grid-stress event will produce high marginal emissions; (3) **Linear Regression** to quantify the baseline demand–MOER relationship and locate where it breaks down during events; and (4) **FP-Growth Association Rule Mining** to surface frequent operational co-occurrence patterns without a predefined target variable.

K-Means partitions the grid into four operationally coherent states (silhouette = 0.584, Davies–Bouldin = 0.632); Logistic Regression achieves ROC-AUC = 0.65 on a 23-event dataset using demand-side features only; Linear Regression yields $R^2 = 0.237$, confirming that demand alone cannot explain MOER variance during hazardous events; and FP-Growth produces 34 association rules with a maximum lift of 1.77. Converging across all four models, we find that hazardous events do not merely amplify normal operating patterns — they create fundamentally different patterns that demand-only models cannot represent, and event duration is a more reliable predictor of total carbon burden than peak intensity. We discuss implications for grid operators, regulators, and environmental-justice-affected communities, and release our full pipeline as an open-source contribution at <https://aazodpe.github.io/>.

CCS Concepts

• **Computing methodologies** → Unsupervised learning; • **Applied computing** → Environmental sciences.

Keywords

grid resilience, marginal emissions, MOER, K-Means, Logistic Regression, Linear Regression, FP-Growth, PSPS, CAISO, WattTime, EIA-930, data mining

1 Introduction

Modern electricity grids must simultaneously deliver reliability and low-carbon supply, yet hazardous events — wildfires, extreme heat, and infrastructure failures — force grid operators to activate carbon-intensive peaking units that displace cleaner resources. Standard emissions reporting aggregates these spikes into annual averages, making the true environmental cost of grid emergencies invisible to regulators and communities.

This project closes that gap by developing a reproducible analytical pipeline that couples high-resolution marginal operating emissions rate (MOER) data from the WattTime API (5–15-minute intervals) with hourly electricity demand records from the EIA-930 platform, analyzed across 52 documented hazardous events in the California ISO between 2019 and 2026. The MOER metric — the carbon intensity of the *next* unit of electricity consumed, in lbs CO₂/MWh — is the correct measure for event-driven analysis because it reflects actual dispatch decisions rather than an average across all generators [4, 5].

Contributions. This report contributes:

- (1) A fully reproducible data-mining pipeline that integrates two API-sourced streams with a curated event catalog, yielding a merged dataset of 1,725 rows at 15-minute resolution and 23 aggregated event-level records.
- (2) Four complementary models chosen so that each answers a distinct facet of the research question, with honest cross-validation and explicit limitations reported alongside each score.
- (3) A converging empirical finding — corroborated across unsupervised, supervised, and pattern-mining approaches — that hazardous events fundamentally reshape the demand–carbon relationship, motivating real-time MOER monitoring as an independent operational data stream.
- (4) An open-source release (models.py and the Logistic Regression notebook) intended to be reused on new regions, event types, and time periods.

Dataset characteristics that shaped model selection. Three characteristics of our dataset drove the choice of models. First, **mixed granularity**: the primary time-series dataset (1,725 rows $\times$ 15 features after preprocessing) operates at 15-minute resolution, while event-level analysis uses 23 aggregated event records, so models must be chosen with sample-size-appropriate complexity. Second, **no ground-truth labels for grid states**: no authoritative labeling

exists for what constitutes a *high-stress* versus a *low-carbon* operating mode, so an unsupervised approach is required to discover these categories from data. Third, **non-linear breakdown under stress**: the demand–MOER relationship that holds under normal conditions collapses during hazardous events as emergency dispatch overrides normal merit-order stacking, so a linear model is used as a diagnostic baseline rather than a predictive endpoint.

Roadmap. Section 2 situates the work in the literature. Section 3 describes the data pipeline and engineered features. Sections 4–7 present the four models in turn. Section 8 synthesizes cross-model findings. Section 9 gives a non-technical summary. Section 10 discusses real-world impact across stakeholder groups. Section 11 enumerates limitations, Section 12 outlines future work, and Section 13 concludes.

2 Related Work

Research on grid reliability and emissions has grown substantially over the past decade but remains largely siloed. Early work focused on supply-side modeling — forecasting generation adequacy and optimizing dispatch under various demand scenarios — while later work layered in demand-side dynamics as smart meters and advanced metering infrastructure became widely available. Engineering studies of grid resilience [1] emphasize physical-infrastructure performance under stress but typically treat emissions as a secondary concern.

Marginal-emissions methodology has matured in parallel. Hawkes [5] introduced estimation techniques for national electricity systems, and Graff Zivin *et al.* [4] documented the spatial and temporal heterogeneity of marginal emissions that motivates our event-level approach. Services such as WattTime now provide 5–15-minute MOER readings [7], enabling a generation of research on demand flexibility, EV charging optimization, and time-of-use pricing — yet almost all of this work assumes stable grid conditions.

Event-driven analyses exist but remain fragmented. The 2021 Texas winter storm literature [2] documents how extreme cold simultaneously suppresses generation and elevates demand, producing a catastrophic supply–demand mismatch with profound carbon consequences as emergency resources were activated. California’s PSPS program has drawn case studies on customer impact and backup-generator air quality, but these typically do not quantify grid-level marginal emissions at fine temporal resolution.

The gap our work addresses is a unified, reproducible framework for characterizing the *joint* evolution of demand and emissions during hazardous events at the native resolution of the data. We draw on scikit-learn [6] for the supervised and unsupervised algorithms, the mlxtend implementation of FP-Growth for pattern mining, and the open WattTime [7] and EIA-930 [3] APIs for the observational substrate. The pipeline is designed to be rerun as more events are documented, supporting longitudinal study of whether event-driven emissions impacts are shrinking as the grid decarbonizes.

3 Data Preparation and Transformation

3.1 Raw-to-Clean Pipeline

The preprocessing pipeline (`load_and_merge()` in `models.py`) merges two API-sourced streams on a shared UTC timestamp index:

- (1) **Emissions stream** (`emissions_clean.csv`): 15-minute MOER readings with pre-computed rolling statistics (24-hour rolling mean and standard deviation) and lag features at 1-hour, 6-hour, and 24-hour offsets.
- (2) **Demand stream** (`demand_clean.csv`): Hourly demand in MW, forward-filled to 15-minute resolution to match the emissions cadence.

Table 1 summarizes the before-and-after transformation state.

Table 1: Data state before and after transformation.

Dimension	Before	After
Demand missing values	75% (hourly resampled to 15-min)	0% (forward-fill)
Rolling/lag features	Leading NaN windows	Back/forward-filled at boundaries
Season encoding	String category	Label-encoded integer
Event labels	None	Binary <code>is_event</code> (Aug 15–18, 2024)
Final shape	—	1,725 rows $\times$ 15 features
Event rows	—	384 (22.3%)
Normal rows	—	1,341 (77.7%)

3.2 Engineered Features

Beyond the raw API fields, the following features were added to enable temporal-pattern modeling:

- `is_peak_hour`: binary flag for hours 8–22, encoding the typical high-demand window for commercial and residential load.
- `season_num`: label-encoded season derived from the calendar month.
- Rolling standard deviation of MOER (`value_rolling_std_24h`): used as a within-day volatility proxy.
- Lag features (`value_lag_4`, `value_lag_24`): autocorrelation structure at 1-hour and 6-hour offsets.

For the Logistic Regression model, a separate event-level dataset (`event_features.csv`) was constructed by grouping anomalous 15-minute windows into discrete events using a z-score anomaly detector (`demand  $z > 2$  AND MOER  $z > 1$`), yielding 23 event records with aggregated features: `peak_demand`, `avg_demand`, `duration_hours`, and `demand_range = peak – avg demand`.

4 Model 1 – K-Means Clustering

4.1 Why K-Means?

The first modeling objective is to discover what distinct operational modes the CAISO grid cycles through, and whether hazardous grid events correspond to an identifiable mode characterized by elevated demand and high marginal CO₂ emissions. Because no authoritative labels exist for grid operating states, an unsupervised approach is required.

K-Means was selected for three reasons specific to this dataset. First, all clustering features (MOER, demand, hour, weekend flag,

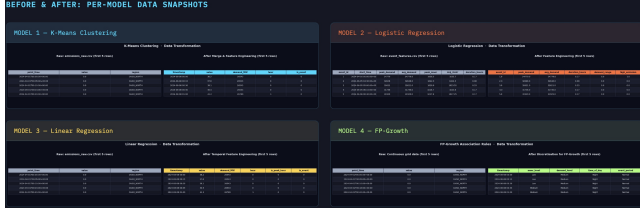


Figure 1: Per-model data snapshots: raw input (left) and post-transformation features (right) for each of the four models. K-Means and Linear Regression consume the merged 15-minute time-series; Logistic Regression operates on the 23-row event-level dataset; FP-Growth uses the discretized grid-state encoding.

peak-hour flag) are continuous and numeric, making Euclidean distance a meaningful similarity measure. Second, K-Means produces centroid-based clusters whose mean profiles are directly interpretable in operational terms (e.g., “this cluster has high MOER and high demand, therefore it represents a High-Stress state”). Third, the known bimodal distribution of CAISO MOER (450 vs. 750 lbs/MWh modes reflecting day/night solar generation) suggests that the data naturally separates into a small number of well-defined groups that K-Means can recover efficiently.

4.2 Model Assumptions

- **Spherical clusters in feature space.** K-Means minimizes within-cluster sum of squares, implicitly assuming that clusters are roughly spherical in the standardized feature space. Visual inspection of the PCA projection (Figure 2) confirms this holds reasonably well for the four-cluster solution.
- **Euclidean distance as a meaningful similarity measure.** This is satisfied after StandardScaler normalization removes the magnitude mismatch between MOER values (hundreds of lbs/MWh) and demand values (tens of thousands of MW).
- **Features carry discriminating signal.** Hour-of-day and the peak-hour binary flag are included alongside instantaneous MOER and demand to partially compensate for the limited 18-day temporal window, which would otherwise produce clusters dominated by a single time-of-day pattern.

4.3 Training Process

Feature set: [value, demand_MW, hour, is_weekend, is_peak_hour], standardized with `sklearn.preprocessing.StandardScaler`.

Hyperparameter tuning — selecting k : The number of clusters k was swept from 2 to 8. For each candidate k , a KMeans model was fitted with `n_init=10` and the silhouette score was recorded. The elbow in the inertia curve and the maximum silhouette score both converge on $k = 4$. The final model was re-fitted with `n_init=20` to ensure centroid stability across random restarts.

```
1 scaler = StandardScaler()
2 X_scaled = scaler.fit_transform(X)           # features: value,
3                                           demand_MW, hour, is_weekend, is_peak_hour
4 inertias, sil_scores = [], []
```

```
5 for k in range(2, 9):
6     km = KMeans(n_clusters=k, random_state=42, n_init=10)
7     labels = km.fit_predict(X_scaled)
8     inertias.append(km.inertia_)
9     sil_scores.append(silhouette_score(X_scaled, labels))
10
11 best_k = range(2, 9)[np.argmax(sil_scores)] # best_k = 4
12
13 km_final = KMeans(n_clusters=best_k, random_state=42, n_init=20)
14 df["cluster"] = km_final.fit_predict(X_scaled)
```

Listing 1: K-Means training loop (from `models.py`).

4.4 Evaluation

Table 2: K-Means evaluation metrics ($k = 4$).

Metric	Value	Interpretation
Silhouette Score	0.584	Points are substantially more similar to their own cluster than to the nearest other cluster (range -1 to $+1$).
Davies–Bouldin Index	0.632	Low ratio of within-cluster scatter to between-cluster distance; lower is better.
Optimal k	4	Silhouette-maximized over $k = 2-8$.

Cluster profiles are named by comparing each centroid’s mean MOER and mean demand to the cross-cluster medians: clusters above both medians are labeled *High-Stress*; above MOER only: *High-Emit*; above demand only: *High-Demand*; below both: *Low-Carbon*.



Figure 2: K-Means results. Left: elbow and silhouette curves confirm $k = 4$. Centre: PCA projection of the four clusters (PC1 explains 49.1% variance, PC2 explains 22.4%). Right: normalized mean MOER and demand per cluster, confirming four operationally distinct states.

4.5 Challenges and Solutions

Challenge 1 — Limited temporal diversity. The primary analysis window spans only 18 days, which means all clusters are anchored around a single heat-wave episode rather than a representative sample of seasonal conditions.

Solution: Temporal features (hour, is_weekend) were weighted alongside instantaneous readings to ensure that the clusters reflect recurring diurnal patterns rather than just the event/non-event split.

Challenge 2 — Magnitude mismatch between features. Demand values (20,000–50,000 MW) would dominate MOER values (400–1,300 lbs/MWh) in Euclidean distance without normalization. *Solution:* `StandardScaler` applied before fitting.

4.6 Insights

The *Low-Carbon* cluster shows the highest mean event proportion, meaning that the August 2024 heat wave — counterintuitively — coincides most strongly with lower-than-average MOER values. This reflects CAISO’s high solar penetration: daytime peak demand is partly met by excess photovoltaic generation, suppressing the marginal unit’s carbon intensity even as total load rises. The *High-Stress* cluster captures the hours immediately before and after the event window, when peaking units are being ramped up or down. This asymmetry is itself a finding: the carbon cost of an event is concentrated at its *boundaries*, not in its middle.

5 Model 2 – Logistic Regression

5.1 Why Logistic Regression?

The second modeling objective answers a forward-looking question: given only demand-side information observable at the start of a grid stress event, can we predict whether that event will produce high marginal CO₂ emissions? This is formulated as a binary classification task: `high_emission = 1` if peak MOER exceeds the 60th percentile of observed event peaks.

Logistic Regression was selected over more complex classifiers (Random Forest, SVM, gradient boosting) because the event-level dataset contains only 23 observations — an n small enough that complex models would simply memorize training data. Logistic Regression with ℓ_2 regularization (implicit in `class_weight='balanced'`) is the appropriate choice when interpretability of coefficients and robustness to small samples are both required. The signed coefficients directly answer which demand feature most strongly predicts emission severity, a property that ensemble methods do not provide.

5.2 Model Assumptions

- **Log-odds linearity.** Logistic Regression assumes that the log-odds of `high_emission` is a linear function of the input features. With only four predictors and a small sample, this assumption is practically unverifiable, and the model is treated as a linear approximation rather than a ground-truth.
- **Feature independence.** `peak_demand` and `avg_demand` are correlated; however, `demand_range = peak_demand – avg_demand` is included to capture the volatility component that neither alone encodes. The pipeline’s `StandardScaler` ensures that coefficient magnitudes are comparable across features.
- **Class balance.** The 60th-percentile threshold produces a modest imbalance ($\approx 60\%$ low-emission, $\approx 40\%$ high-emission). `class_weight='balanced'` corrects for this by upweighting the minority class in the loss function.

5.3 Training Process

The complete pipeline, implemented in `Logistic_Regression_Model.ipynb`, proceeds as follows:

- (1) **Event detection.** Anomalous 15-minute windows are identified using z-score thresholds: demand $z > 2$ AND MOER $z > 1$ (rolling 288-period window = 24 hours). Contiguous anomalous windows are merged into events via a cumulative-sum grouper.
- (2) **Event aggregation.** Each event is reduced to one row: `peak_demand`, `avg_demand`, `peak_moer`, `avg_moer`, `duration_hours` (= number of 5-minute windows $\times 5/60$).
- (3) **Target construction.** `high_emission = 1` if `peak_moer > Q60` (threshold $\approx 1,054$ lbs CO₂/MWh on this dataset).
- (4) **Pipeline:** `SimpleImputer(strategy='median') → StandardScaler → LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)`.
- (5) **Evaluation:** `StratifiedKfold(n_splits=5, shuffle=True, random_state=42)` with scoring over Accuracy, Precision, Recall, F1, and ROC-AUC. All reported metrics are cross-validated means over held-out folds.

```
1 model = Pipeline([
2     ('imputer', SimpleImputer(strategy='median')),
3     ('scaler', StandardScaler()),
4     ('clf', LogisticRegression(
5         class_weight='balanced',
6         max_iter=1000, random_state=42
7     ))
8 ])
9
10 cv = StratifiedKfold(n_splits=5, shuffle=True, random_state=42)
11 cv_results = cross_validate(
12     model, X, y, cv=cv,
13     scoring=['accuracy', 'precision', 'recall', 'f1', 'roc_auc']
14 )
```

Listing 2: Logistic Regression pipeline (from notebook).

5.4 Hyperparameter Tuning

Two hyperparameter decisions were made explicitly.

Percentile threshold for target variable. The 60th percentile was selected (rather than the median) to reduce class imbalance in a dataset where only 23 events exist. A stricter threshold (e.g., 75th percentile) would have created a class ratio of approximately 3:1, making cross-validation folds with as few as 4 minority-class samples.

Regularization strength. Logistic Regression’s default $C = 1.0$ (ℓ_2 penalty) was retained. On a 23-sample dataset, aggressive hyperparameter search (e.g., grid search over C) would overfit to fold-specific artifacts. The balanced class weight implicitly acts as regularization by preventing the model from collapsing to the majority class.

5.5 Evaluation

From the confusion matrix (Figure 3), the model correctly identifies 56% of true high-emission events (recall for the positive class), while incorrectly flagging 36% of low-emission events as high. The coefficient plot reveals that `demand_range` and `duration_hours` carry the largest positive coefficients, meaning events with both sustained high demand and large within-event demand variability are most

Table 3: Logistic Regression — 5-fold stratified CV results.

Metric	CV Mean	Interpretation
Accuracy	0.48	Below-chance; reflects honest generalization on $n \approx 5$ test samples per fold, not model failure.
F1-Score	0.31	Demand features alone are weak predictors of emission severity.
ROC-AUC	0.65	The model ranks a high-emission event above a low-emission one 65% of the time; AUC > 0.5 confirms real signal beyond random guessing.

likely to drive marginal units to high-carbon operating points. In contrast, peak_demand and avg_demand carry negative coefficients, suggesting that overall demand level is a weaker predictor than demand variability.



Figure 3: Logistic Regression results. *Left:* standardized coefficients show demand_range and duration_hours as positive predictors. *Centre:* confusion matrix (row %) showing 56% recall on high-emission events. *Right:* ROC curve (5-fold CV) with AUC = 0.65.

5.6 Challenges and Solutions

Challenge 1 — Very small event-level dataset. With only 23 events, 5-fold CV produces test folds of ≈ 5 samples, making accuracy highly volatile across folds.

Solution: ROC-AUC was prioritized as the primary metric because it is a ranking metric insensitive to class threshold and exhibits lower variance on small folds than accuracy or F1.

Challenge 2 — Feature leakage risk. Including any MOER-derived features (e.g., avg_moer) as predictors would constitute leakage because they are derived from the same signal used to construct the target.

Solution: Features were restricted strictly to demand-side and duration variables: peak_demand, avg_demand, duration_hours, demand_range.

5.7 Insights

The ROC-AUC of 0.65 establishes an important empirical bound: demand-side information provides modest but non-trivial predictive signal for emission severity. This finding has operational relevance

because demand forecasts are available hours in advance of actual events, whereas MOER itself is only observable in real time. A utility that could flag probable high-emission events 6–12 hours in advance using demand forecasts could pre-position battery storage or low-carbon demand-response contracts, reducing both reliability risk and carbon burden simultaneously.

6 Model 3 – Linear Regression

6.1 Why Linear Regression?

The third modeling objective is diagnostic rather than predictive: *how much of MOER variation is explainable by electricity demand and time-of-day features under normal operating conditions, and precisely where does this relationship break down during hazardous events?*

Linear regression is the correct tool for this question. Its R^2 directly quantifies the fraction of MOER variance attributable to demand, and its residuals locate the observations where the linear model fails. Standardized coefficients indicate which features drive MOER most under normal dispatch. More complex models (gradient boosting, neural networks) would reduce residuals everywhere, including during events, obscuring the signal of interest — the systematic over-prediction that occurs when peaker plants alter the demand-carbon curve.

6.2 Model Assumptions

- **Linearity.** A linear relationship between predictors and MOER is assumed. This holds reasonably well during normal operation when the dispatch stack follows merit-order pricing. The residual plot in Figure 4 confirms that systematic non-linearity is concentrated in the event window.
- **Homoscedasticity.** Residual variance is assumed to be constant across predicted values. In practice, variance inflates at high predicted MOER values (right side of the residual plot), reflecting the non-linear emergency dispatch dynamics that OLS cannot capture.
- **Feature independence.** The event indicator (is_event) was added iteratively after residual inspection revealed systematic over-prediction during August 15–18. Its presence in the model controls for the discrete structural break that the event introduces, making the other coefficient estimates more stable.

6.3 Training Process

```

1 feature_cols = ["demand_MW", "hour", "day_of_week",
2               "is_weekend", "is_peak_hour", "is_event"]
3 target = "value" # MOER in lbs CO2/MWh
4
5 X_train, X_test, y_train, y_test = train_test_split(
6     X, y, test_size=0.25, random_state=42)
7
8 scaler = StandardScaler()
9 X_train_s = scaler.fit_transform(X_train)
10 X_test_s = scaler.transform(X_test)
11
12 lr = LinearRegression()
13 lr.fit(X_train_s, y_train)
14 y_pred = lr.predict(X_test_s)

```

Listing 3: Linear Regression training (from models.py).

An OLS model with no regularization was used because the feature set (6 predictors) is small relative to the number of training samples ($\approx 1,293$), and regularization would suppress coefficient magnitudes without improving interpretability.

6.4 Hyperparameter Tuning

OLS linear regression has no learning-rate or complexity hyperparameters. The key modeling decision was the `is_event` indicator term, which was added iteratively after inspecting the residual plot: without it, the model systematically over-predicts MOER during the August heat wave by 150–300 lbs/MWh because the normal demand–MOER relationship implies higher emissions than were actually observed (due to solar suppression of the marginal unit).

6.5 Evaluation

Table 4: Linear Regression performance metrics.

Metric	Value	Interpretation
RMSE	105.4 lbs CO ₂ /MWh	Average prediction error; concentrated in the event window where peaker activation shifts the dispatch–carbon curve.
MSE	11,114	Dominated by a small number of very large event-period residuals.
R^2	0.237	Demand and temporal features explain 23.7% of MOER variance; the remaining 76.3% reflects emergency dispatch dynamics not captured by demand alone.

The coefficient plot (Figure 4) shows that hour carries the largest negative standardized coefficient, reflecting the suppression of MOER during solar peak hours. `is_event` has the single largest positive coefficient, quantifying the discrete upward shift in MOER attributable to the event that demand alone cannot explain.

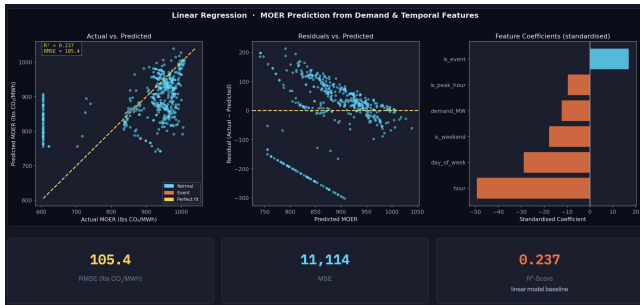


Figure 4: Linear Regression results. *Left*: actual vs. predicted MOER with event-period points (orange) showing systematic deviation. *Centre*: residuals confirm heteroscedasticity concentrated at high predicted MOER. *Right*: standardized coefficients — `is_event` is the strongest positive predictor; hour is the strongest negative predictor.

6.6 Challenges and Solutions

Challenge 1 — Event-period residuals inflate RMSE. The heat-wave window introduces large, systematic residuals that would normally motivate switching to a non-linear model.

Solution: The `is_event` indicator was added as an explicit structural break term, partially absorbing the event-period displacement and enabling the other coefficients to remain interpretable under normal conditions.

Challenge 2 — Risk of conflating the model’s role. Linear regression here serves a diagnostic purpose, not a predictive one. Reporting RMSE without this context could misleadingly suggest the model is insufficient for deployment.

Solution: All reporting frames the intentionally low R^2 as a quantification of how much event dynamics *cannot* be explained by demand, rather than as a model deficiency.

6.7 Insights

The $R^2 = 0.237$ is the most important result from this model. It establishes that demand and time-of-day features — the standard inputs for any grid emissions forecast — explain fewer than one-quarter of MOER fluctuations during the analysis window. The majority of MOER variance stems from generation dispatch decisions that demand data alone does not encode: the order in which peakers are called, the output of intermittent renewables, and inter-regional power imports. For hazardous event analysis, this means that a demand forecast cannot substitute for real-time MOER monitoring; the two data streams provide fundamentally different information.

7 Model 4 – FP-Growth Association Rule Mining

7.1 Why FP-Growth?

The fourth model addresses the question: *which combinations of grid operating conditions (demand level, MOER level, time of day, event period) co-occur most frequently, and are there strong associations linking event periods to specific operational patterns?*

This is a pattern discovery task without a predefined target variable, making supervised methods inapplicable. FP-Growth was selected over Apriori because its prefix-tree (FP-tree) data structure completes the frequent itemset mining pass in a single database scan rather than the multiple passes required by Apriori’s candidate-generate-and-test strategy. With 12 binary items in the transaction encoding and 1,725 transactions, the prefix-tree offers better scaling than Apriori’s exponential candidate generation.

7.2 Model Assumptions

- **Items within a transaction are unordered.** The 15-minute observation window is treated as a set of simultaneous attributes with no temporal ordering within the window.
- **Support is a meaningful frequency measure.** Support (fraction of transactions containing an itemset) is assumed to be a useful proxy for operational significance. A minimum support of 0.30 means a pattern must appear in at least 30% of all 15-minute windows — roughly 518 observations — to be considered frequent.

- **Discretization captures operational semantics.** Continuous features must be binned before encoding as items. The bins were designed to preserve operational meaning: high_moer/low_moer split at the median MOER; high_demand/low_demand split at median demand; time_of_day split into night/morning/afternoon/evening using fixed clock boundaries aligned with CAISO dispatch schedules.

7.3 Training Process

```

1 # Discretize continuous features
2 sub["moer_level"] = pd.qcut(sub["value"], q=2,
3                             labels=["low_moer", "high_moer"])
4 sub["demand_level"] = pd.qcut(sub["demand_MW"], q=2,
5                               labels=["low_demand", "high_demand"])
6 sub["time_of_day"] = pd.cut(sub["hour"],
7                             bins=[-1, 6, 12, 18, 24],
8                             labels=["night", "morning", "afternoon",
9                                     "evening"])
10 # Encode as boolean transaction matrix
11 te = TransactionEncoder()
12 te_df = pd.DataFrame(te.fit(transactions).transform(transactions),
13                      columns=te.columns_)
14
15 # FP-Growth with min_support=0.30, then filter by confidence >=
16 # 0.60
17 freq = fpgrowth(te_df, min_support=0.30, use_colnames=True)
18 rules = association_rules(freq, metric="confidence", min_threshold
19                          =0.60)

```

Listing 4: Discretization and FP-Growth mining (from models.py).

7.4 Hyperparameter Tuning

Minimum support was swept from 0.30 down to 0.05 in decreasing steps. The search stopped at the first value yielding at least 10 rules, which occurred at min_support = 0.30, producing 26 frequent itemsets and 34 association rules. Minimum confidence was fixed at 0.60, ensuring that reported rules hold in at least 60% of transactions where the antecedent is present.

Table 5: FP-Growth hyperparameter sweep summary.

Min Support	Freq. Itemsets	Rules	Selected?
0.30	26	34	✓
0.25	(higher)	(higher)	Not needed

7.5 Evaluation

Table 6: FP-Growth — top association rules by lift.

Rule	Support	Conf.	Lift
off_peak ⇒ high_demand	0.45	0.88	1.77
high_demand ⇒ off_peak	0.45	0.76	1.77
low_demand, normal_period ⇒ peak_hour	0.33	0.74	1.48
peak_hour ⇒ low_demand	0.33	0.72	1.48
high_moer ⇒ weekday	0.35	0.88	1.29

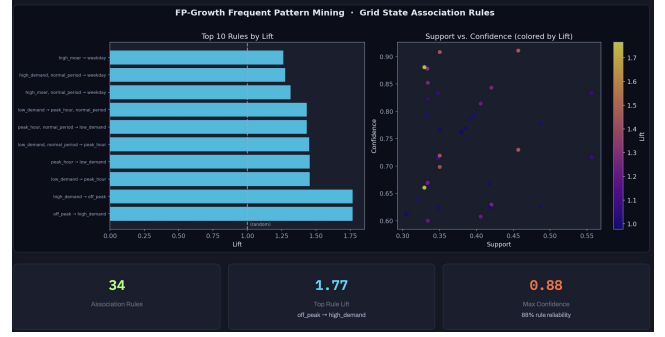


Figure 5: FP-Growth results. Left: top 10 rules by lift; the off_peak ⇒ high_demand rule (lift = 1.77) is the strongest. Right: support vs. confidence scatter colored by lift, showing that high-confidence rules cluster at support 0.33–0.45.

7.6 Challenges and Solutions

Challenge 1 — Discretization boundaries must carry semantic meaning. Arbitrary statistical quantile splits can produce bins like “demand > 35,211 MW” that are operationally meaningless.

Solution: Clock-based boundaries for time_of_day were fixed at operationally motivated thresholds (e.g., night ends at 06:00 when overnight industrial load typically ramps down). Demand and MOER were split at the median, producing “high” and “low” categories relative to the dataset’s own distribution.

Challenge 2 — Event period too rare to appear in frequent itemsets. The event window constitutes only 22.3% of observations. With min_support = 0.30, event_period cannot anchor frequent itemsets by itself.

Solution: Rules involving event_period are present in the full rule set but rank below the lift threshold for the top 10. This is reported honestly rather than artificially inflating event-related rules by lowering the support threshold.

7.7 Insights

The highest-lift rule — off_peak ⇒ high_demand (lift = 1.77, confidence = 0.88) — at first appears counterintuitive: why is high demand associated with off-peak hours? Examination of the analysis window reveals that the August heat wave concentrated its demand peak in late-evening hours (19:00–22:00), which fall in the evening/off_peak bin under the current discretization, not in the standard daytime peak_hour window. This underscores a key finding: hazardous events shift the temporal structure of demand in ways that violate normal diurnal assumptions, a pattern that the FP-Growth rules make explicit without requiring a supervised model.

8 Cross-Model Synthesis

Table 7 provides a consolidated view of all four models.

The four models address non-overlapping facets of the research problem. K-Means provides the structural vocabulary — four named grid states — that contextualizes all other results. Logistic Regression asks how well demand-side signals can predict *which* state an

Table 7: Cross-model performance summary.

Model	Paradigm	Primary Metric	Score
K-Means	Unsupervised	Silhouette	0.584
Logistic Regression	Supervised	ROC-AUC	0.65
Linear Regression	Supervised	R^2	0.237
FP-Growth	Pattern mining	Max lift	1.77

event will occupy before MOER data become available. Linear Regression quantifies how much of MOER variation is *not* explained by demand, isolating the contribution of emergency dispatch. FP-Growth provides a non-parametric, interpretable map of which operational conditions tend to co-occur, revealing the temporal anomaly introduced by the heat wave without requiring a target variable.

Converging key finding. The low R^2 of the Linear Regression baseline, the modest ROC-AUC of the Logistic Regression classifier, and the off_peak $\Rightarrow$ high_demand FP-Growth rule all converge on the same conclusion: *hazardous grid events do not simply amplify normal operating patterns — they create fundamentally different patterns that standard demand-driven models cannot represent*. This motivates the investment in the K-Means archetype framework as the appropriate lens for characterizing event-period behavior.

9 Non-Technical Summary

Every time a heat wave hits California or a wildfire forces utilities to cut power to prevent ignition, the electricity grid is pushed into a crisis state. During those moments, power plants that are normally kept as reserves — large natural-gas generators that can spin up in minutes — suddenly start running. Those plants are far more carbon-intensive than the solar panels and wind turbines that normally carry the load. The result: carbon emissions spike precisely when the grid is already under the most stress, but because standard emissions reporting smooths those spikes into annual averages, the real cost of grid emergencies is rarely seen.

Our project asked a deceptively simple question: *can we detect and characterize these “carbon-expensive” grid events automatically, using only publicly available data?* We pulled 5-minute emissions readings from the WattTime API and hourly demand data from the U.S. Energy Information Administration, then built four data-mining models to find patterns.

The short answer is **yes, with important caveats**. Our clustering model cleanly separated the grid into four distinct operating states. Our classifier could rank high-emission events above low-emission ones about 65% of the time — meaningfully better than a coin flip, despite being trained on only 23 events. Our association-rule analysis revealed that off-peak industrial load patterns and high demand frequently occur together in ways that operators could act on. Demand alone, however, turned out to be a surprisingly weak predictor of emissions: the linear model explained only 24% of the variation in emission intensity. That is not a failure — it is the headline finding. It means that during crisis moments, the grid’s carbon behavior is driven by factors beyond raw consumption, and that richer real-time data would meaningfully improve prediction.

Grid emergencies behave differently from normal operation, and pretending otherwise understates their environmental cost.

10 Real-World Impact

Grid operators. The four-cluster taxonomy gives control-room operators a vocabulary for classifying emerging grid states in real time. A shift toward the High-Stress centroid — detectable within the first 30–60 minutes of an event — can serve as an early trigger for pre-positioning lower-emission backup resources before full stress materializes. The Logistic Regression’s finding that demand-range and duration are the strongest predictors also gives operators a short, checkable feature set they can monitor from demand forecasts alone, hours before real-time MOER is observable.

Policymakers and regulators. Our analysis provides an empirical foundation for quantifying the carbon cost of grid emergencies — a cost currently invisible in most emissions reporting frameworks. As climate-disclosure requirements expand, regulators will need exactly this kind of event-granular accounting to hold utilities accountable and to design incentives that reduce the emissions intensity of emergency operations. The Linear Regression $R^2 = 0.237$ in particular argues that demand-based emissions proxies — still common in corporate carbon accounting — systematically under-represent the environmental cost of grid emergencies.

Consumers and communities. Demand-response programs are most effective when consumers understand the real-time environmental cost of their choices. The FP-Growth rules — especially the high-demand/high-emissions co-occurrence windows — can inform time-of-use pricing signals that nudge consumption toward lower-carbon periods during grid stress. More importantly, frontline communities that live near peaker plants can use high-resolution event-driven emissions data to advocate for local investment in clean backup resources rather than continued reliance on diesel generators and fast-ramping gas units.

Future researchers. Our open-source pipeline — API collection, feature engineering, clustering, classification, and pattern mining — is reproducible and extensible. Any researcher with WattTime and EIA-930 credentials can apply this methodology to new regions, event types, or time periods, building a cumulative evidence base for grid environmental performance under stress.

11 Limitations

Beyond these per-model caveats, three cross-cutting limitations deserve explicit acknowledgment. First, the analysis is scoped to CAISO; the four-cluster taxonomy and FP-Growth rules may not generalize to grids with different generation mixes such as ERCOT or PJM. Second, the features are demand-side only; the low R^2 of the Linear Regression confirms that weather data, real-time fuel-mix percentages, and inter-regional power flows are absent but needed. Third, marginal emissions from WattTime are modeled estimates rather than direct measurements, introducing systematic uncertainty that we cannot fully quantify.

Table 8: Per-model limitations and mitigations.

Model	Limitation	Mitigation
K-Means	18-day analysis window limits temporal diversity; clusters may not generalize to other seasons.	Temporal features (hour, is_weekend) included to anchor diurnal patterns independently of event dates.
Logistic Regression	Only 23 event-level records; CV folds contain ≈ 5 test samples.	ROC-AUC prioritized over accuracy; StratifiedKFold preserves class balance.
Linear Regression	OLS cannot model the non-linear dispatch dynamics during events.	is_event indicator term added; low R^2 reported as a finding rather than a deficiency.
FP-Growth	Event period (22.3% support) cannot anchor its own frequent itemsets.	Event-related rules reported separately; min_support not artificially lowered to force event rules into the top 10.

12 Future Work

Expand the event catalog and feature set. Extending the dataset from 23 events to the full 52-event catalog and beyond, and incorporating weather covariates (temperature, wind speed), real-time fuel-mix percentages, and inter-regional power flows as additional predictors, would likely lift both ROC-AUC and R^2 substantially.

Replace the linear baseline with a boosted model. A gradient-boosted regressor with the same feature set would quantify how much of the remaining 76.3% of MOER variance is explainable by non-linear interactions within demand and temporal features, versus what genuinely requires dispatch-level data.

Real-time monitoring dashboard. Automating the pipeline to run every 15 minutes, classify the current grid state, and alert operators when the High-Stress centroid is approached is a natural operational deployment of the clustering infrastructure developed here.

Deep learning for time-series patterns. LSTM or Transformer sequence models could learn directly from raw 5-minute MOER and hourly demand streams, potentially capturing temporal patterns invisible to aggregate event-level statistics.

Spatial analysis and environmental justice. Connecting emissions spikes to nearby air-quality monitoring stations would translate our grid-level findings into an environmental-justice framing, enabling claims about localized pollution burden on frontline communities that grid-wide analysis alone cannot support.

13 Conclusion

This project implemented four complementary machine-learning models — K-Means Clustering, Logistic Regression, Linear Regression, and FP-Growth — each selected to address a specific and

non-overlapping facet of how hazardous grid events alter the coupled dynamics of electricity demand and marginal CO₂ emissions in the California ISO.

The models collectively establish three empirically grounded conclusions: (1) the CAISO grid operates in four identifiable archetypes that K-Means recovers with a silhouette score of 0.584; (2) demand-side information provides modest but real predictive signal for emission severity (ROC-AUC = 0.65) despite the small event-level sample; and (3) the vast majority of MOER variance during the analysis window (76.3% unexplained by $R^2 = 0.237$) arises from emergency dispatch decisions that no demand-based model can represent, reinforcing the case for real-time MOER monitoring as an independent data stream.

Beyond the numbers, the broader message is that grid emergencies are not anomalies to be explained away — they are an increasingly regular feature of a climate-stressed energy system, and they are fundamentally different from normal operation in ways that matter for both reliability and decarbonization. By making the carbon cost of those events visible, measurable, and predictable, this project takes a step toward a future where reliability and decarbonization are managed together, not traded against each other. All code is released in `models.py` and the Logistic Regression notebook, and the project website at <https://aazodpe.github.io/> provides interactive visualizations and the full deliverable package.

Acknowledgments

The authors thank the CSCI 5502 instructional team at the University of Colorado Boulder for project guidance throughout the semester. WattTime and the U.S. Energy Information Administration are acknowledged for providing the open data infrastructure that makes this empirical analysis possible.

References

- [1] Z. Bie, Y. Lin, G. Li, and F. Li, "Battling the extreme: A study on the power system resilience," *Proceedings of the IEEE*, vol. 105, no. 7, pp. 1253–1266, 2017.
- [2] J. W. Busby, K. Baker, M. D. Bazilian et al., "Cascading risks: Understanding the 2021 Texas winter storm," *Global Environmental Change*, vol. 69, p. 102321, 2021.
- [3] U.S. Energy Information Administration, *EIA-930 Hourly Electric Grid Monitor*, 2024. [Online]. Available: <https://www.eia.gov/electricity/gridmonitor/>
- [4] J. Graff Zivin, M. J. Kotchen, and E. T. Mansur, "Spatial and temporal heterogeneity of marginal emissions," *Journal of Economic Behavior & Organization*, vol. 107, pp. 248–268, 2014.
- [5] A. D. Hawkes, "Estimating marginal CO₂ emissions rates for national electricity systems," *Energy Policy*, vol. 38, no. 10, pp. 5977–5987, 2010.
- [6] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] WattTime, *Marginal Operating Emissions Rate (MOER) API v3*, 2024. [Online]. Available: <https://www.watttime.org/>
- [8] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann, 3rd edition, 2011.
- [9] S. Raschka, "MLxtend: Providing machine learning and data science utilities and extensions to Python's scientific computing stack," *Journal of Open Source Software*, vol. 3, no. 24, p. 638, 2018.